

Spam Email Detection Using Machine Learning

A Comparative Analysis of Classification Algorithms

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22 November, 2025

Agenda

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- 2 Dataset & EDA
- 3 Results & Analysis
- 4 Discussion
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Problem Statement

The Challenge

Unsolicited and potentially harmful spam messages pose a significant challenge in digital communication, requiring automated detection systems.

Project Goal

Develop an efficient machine learning-based spam detection system capable of automatically classifying messages as **spam** or **ham**.

Key Objectives:

- Design accurate ML model for spam detection
- Apply NLP techniques for text preprocessing
- Evaluate model performance using classification metrics
- Develop lightweight real-time application

Data Processing

- Data Collection (spam.csv)
- Data Cleaning
- Exploratory Data Analysis

Text Preprocessing

- Lowercasing
- Tokenization
- Stopword Removal
- Stemming

Feature Engineering

- TF-IDF Vectorization

Model Building

- 8 ML Algorithms
- Hyperparameter Tuning

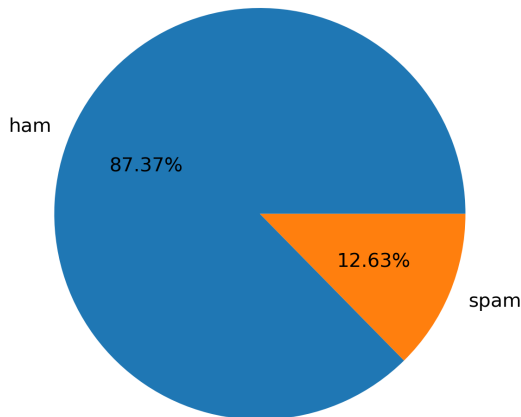
Evaluation

- Accuracy, Precision
- Recall, F1-Score

Dataset Overview

Attribute	Description
Total Messages	5,572
Ham Messages	4,825 (87.37%)
Spam Messages	747 (12.63%)
Features	Message Text, Label
File Format	CSV
Source	Public Spam Collection

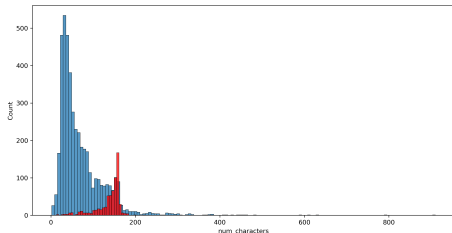
Class Distribution



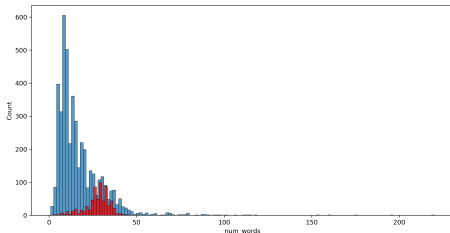
Imbalanced dataset: 87.37% Ham vs 12.63% Spam

Message Length Analysis

Histogram of Spam characters Vs Ham characters

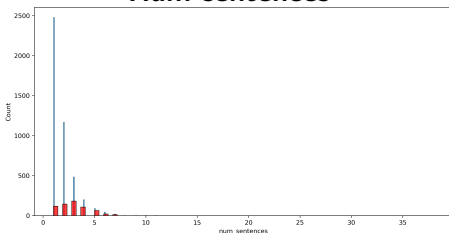


Histogram of Spam words Vs Ham words

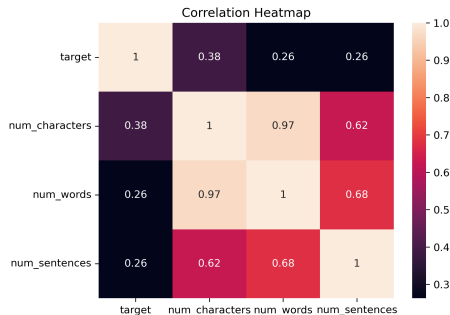


Additional EDA Insights

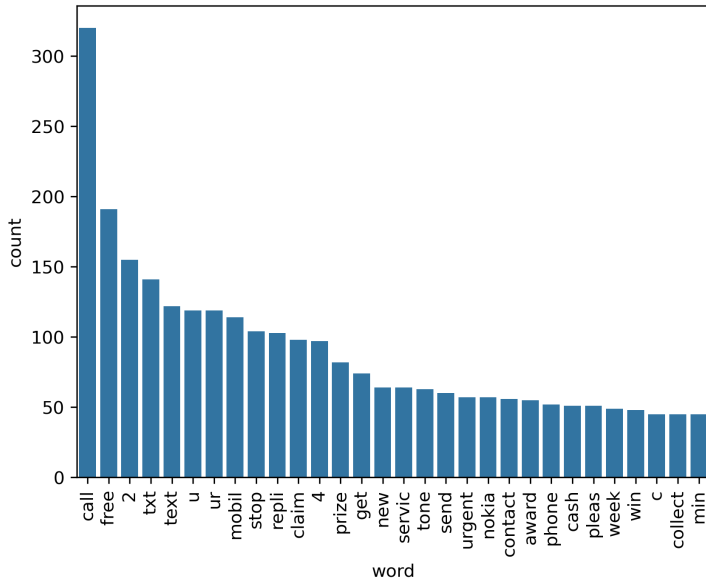
Histogram of Spam sentences Vs Ham sentences



Feature Correlation



Spam Word Analysis



Accuracy

- Overall correctness

Precision

- $$\frac{\text{True Positive(TP)}}{\text{True Positive(TP)} + \text{False Positive(FP)}}$$

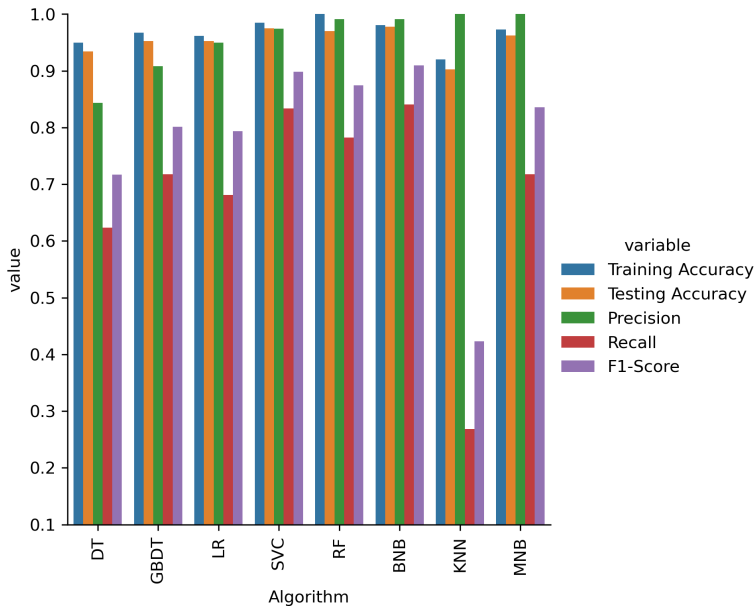
Recall

- $$\frac{\text{True Positive(TP)}}{\text{True Positive(TP)} + \text{False Negative(FN)}}$$

F1-Score

- Balanced measure
- Harmonic mean of precision and recall
- $$\frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$$

Model Performance Comparison



Performance Summary

Model	Accuracy	Precision	Recall	F1-Score
Bernoulli Naïve Bayes	0.977	0.991	0.84	0.909
Support Vector Machine	0.975	0.974	0.833	0.898
Logistic Regression	0.952	0.949	0.681	0.793
Gradient Boosting	0.952	0.908	0.717	0.801
Multinomial Naïve Bayes	0.962	1.0	0.717	0.835
Random Forest	0.97	0.991	0.782	0.874
Decision Tree	0.934	0.843	0.623	0.716
K-Nearest Neighbors	0.902	1.0	0.268	0.422

Key Finding

Bernoulli Naive Bayes achieved the best performance with excellent balance between precision and recall

Discussion & Insights

Best Performing Model

Bernoulli Naive Bayes

- Highest accuracy: **0.977**
- Excellent precision and recall: **0.991** and **0.84** respectively
- Well-suited for high-dimensional text data

Other Strong Performers

- **Support Vector Machine (SVM): 97.5%** accuracy
- **Random Forest: 97.0%** accuracy
- Both showed competitive performance

Success Factors

Effective text preprocessing + TF-IDF features + Appropriate model selection

Conclusion

Achievements

- Successfully developed high-accuracy spam detection system
- **Bernoulli Naive Bayes** achieved **97.7%** accuracy with balanced performance
- Demonstrated effectiveness of NLP + ML for text classification

Key Contributions

- Comprehensive comparison of 8 ML algorithms
- Effective text preprocessing pipeline
- Practical real-time application potential

Future Work

- Address class imbalance (SMOTE)
- Experiment with deep learning (LSTMs)
- Expand dataset and features

References



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Thank You!